Creating Arrays in MATLAB
Lab 03
16/3/2022

Continuing from last week…

Get ready!

1. Create a folder of your own choice
2. Copy the file gasprices.csv to it
3. Start Matlab and navigate to your folder
4. Use Matlab import date to import the gas prices (as you did last week)
5. Verify that the data has imported successfully

Creating Vectors

Vectors are one dimensional lists of data.

You can create a vector by typing its contents directly, for example:

year = [2018 2019 2020 2021 2022]

will create a raw vector

popsize = [100 ; 200; 300; 400; 500]

will create a column vector

Use the ’ operator to convert between raw and column format (transpose):

Pop = popsize’

Creating vectors

Typing the contents of a variable will soon become stale and time consuming if the data is large. You can use the colon operator (:) to create a uniform list of data

yr = 1990:2008

Will create a list of years from 1990 to 2008 with an increment of 1. If you want a different increment, like 5, use:

yr = 1990:5:2008

If you don't know the spacing but you know the number of items in the vector, use linspace(first_element, last_element, number_of_elements)

yr= linspace(1990,2008,10)

Creating Matrices

- A matrix can be entered to Matlab directly raw by raw with each raw separated by a semicolon

e.g. to enter

$$a = \begin{matrix} 3 & 7 & 9 \\ 2 & 4 & 6 \\ 1 & 0 & 3 \end{matrix}$$

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>> a = [3 7 9; 2 4 6; 1 0 3]
```

Let’s practice!

Generate the following matrices using Matlab

$$b = \begin{bmatrix} 1 & 3 \\ 6 & 7 \\ 3 & 2 \end{bmatrix}$$

$$c = \begin{vmatrix} 1 & 5 & 7 \\ 1 & 2 & 3 \end{vmatrix}$$

Some Matrix generating commands

zeros(n,m) % use to generate a matrix of all zeros of size n by m

d = zeros(3,2)

ones(n,m) % use to generate a matrix of all ones of size n by m

e = ones(1,5)

rand(n,m) % use to generate a matrix of all equally-likely random numbers (from 0 to 1) of size n by m

f = rand(3,3)

Matrix Operation

For element by element operation add a dot (.) before the operator. For example:

X = France .* Germany

Y = Mexico ./ USA

For matrix multiplication and division, omit the (.)

Z = France * Germany’

G = inv(f)

Matrix Concatenation and assignment

You can create a matrix by concatenating two, or more, variables together

american_gas = [USA Canada Mexico]

To get the matrix index where the year is 1998 use:

Year == 1998

Now you can use this to find the gas price in any country at that year:

Japan(Year==1998)

To assign a new value to that year

Japan(Year==1998) = 3

Solving a system of linear Equations

Use Matlab to solve the following linear Equations:

$$\begin{bmatrix} 2x + y + 3z = 1 \\ 5y + 5z = 2 \\ 5y + 9z = 2 \end{bmatrix}$$

```
>> a = [2  1  3; 0  5  5;  0  5  9];
>> b = [1  2  2]';
>> x = a\b
x = 0.3  0.4  0
```

Note the special (\ ) operator is used because it is faster that inverting the matrix using inv(a) * b

Let’s Practice!

Use Matlab to solve the following linear Equations:

$$\begin{bmatrix} 3x + 6y - z = -10 \\ 2x + y \ - 5z = -8 \\ 6x - 3y + 3z = 0 \end{bmatrix}$$

Answer

-1.0000 -1.0000 1.0000

Your Home work

1- solve the following system of linear equations:

$$\begin{bmatrix} 4x - 2y + 6z = 8 \\ 2x + 8y + 2z = 4 \\ 6x + 10y + 3z = 0 \end{bmatrix}$$

Your Home work

2- An electric circuit can be described by the following linear equations, solve for i_1 to i_4 using Matlab

$$\begin{bmatrix} -(R_1+R_2+R_3) & R_2 & R_3 & 0 \\ R_2 & -(R_2+R_4+R_5+R_7) & R_4 & R_7 \\ R_3 & R_4 & -(R_3+R_4+R_6) & R_6 \\ 0 & R_7 & R_6 & -(R_6+R_7+R_8) \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} -V_1 \\ 0 \\ V_2 \\ -V_3 \end{bmatrix}$$

$$V_1 = 20\text{ V},\ V_2 = 12\text{ V},\ V_3 = 40\text{ V}$$
$$R_1 = 18\,\Omega,\ R_2 = 10\,\Omega,\ R_3 = 16\,\Omega$$
$$R_4 = 6\,\Omega,\ R_5 = 15\,\Omega,\ R_6 = 8\,\Omega$$
$$R_7 = 12\,\Omega,\ R_8 = 14\,\Omega$$